

MATH503



4.1 Binomial Distribution

Binomial Distribution : a look ahead

- A statistical tool used to find the probability of the number of successes that are likely to occur in a sequence of independent trials
- BINS (Conditions for use that must be met!)
 - Binary : trials can only have 2 possible outcomes (success/failure; yes,/no; true/false)
 - Independent : Outcome of any given trial has no effect on subsequent trials
 - Number : there must be a fixed number of trials
 - Same : the probability of success / failure must be the same (fixed) for all trials
- Let N = number of trials; p = probability of success ; $q = 1 - p$ = probability of failure then
 - Mean : expected number of successes in N trials is $\mu = Np$
 - Variance : measure of spread of the distribution is $\sigma^2 = Np(1 - p)$
- When N is very large the Binomial distribution is well approximated by the Normal distribution (the subject of the week 7 lecture)

Last week: Descriptive statistics

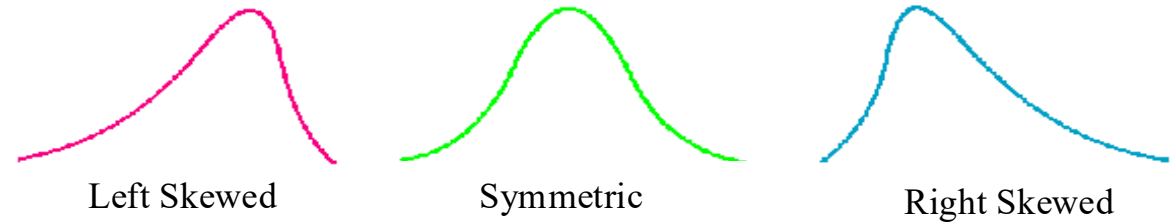
- **Measures of central tendency**

Averages: mean, median, mode

- **Measures of spread (variability)**

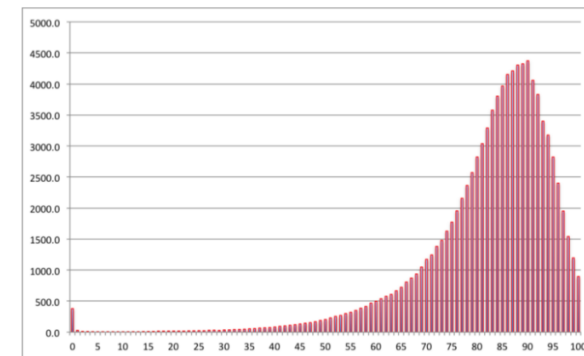
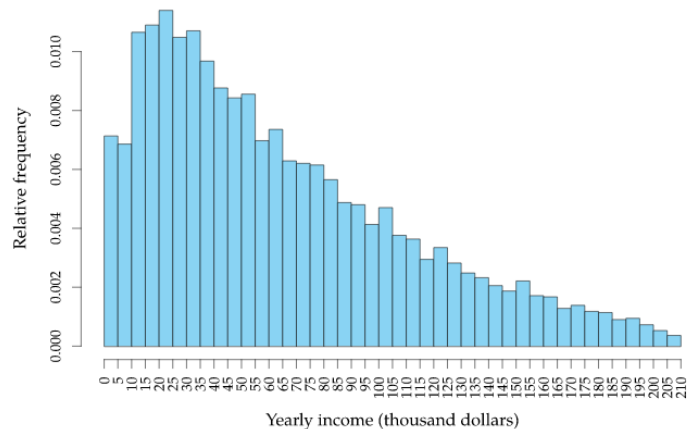
Spread of population: variance, standard deviation, range, maximum, minimum, lower quartile, upper quartile, interquartile range

Shape of distribution



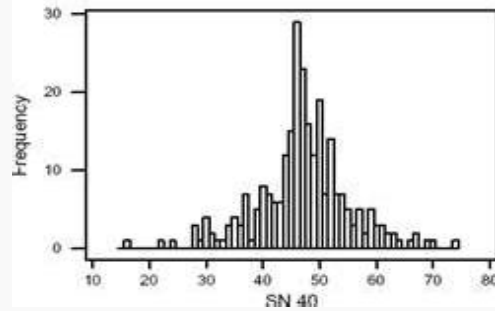
- **right skewed** have a long **tail that extends to the right**
Tail is stretched to the right / in the direction of the **positive** horizontal axis
- **left skewed** have a long **tail that extends to the left**
Tail is stretched to the right / in the direction of the **negative** horizontal axis

Skew definitions are based on what the tail is doing!



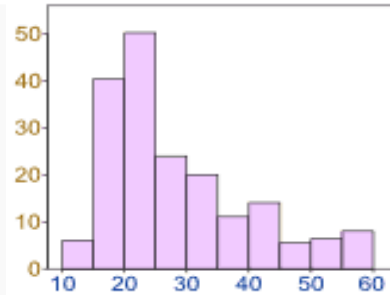
Here the distribution tells that most of the people die at the age of near to 90.

1.



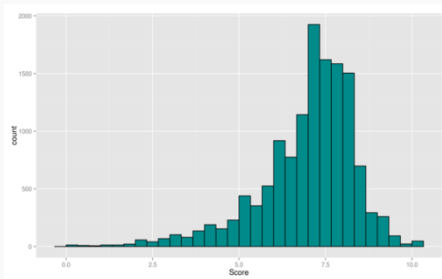
A. Right skewed, unimodal distribution

2.



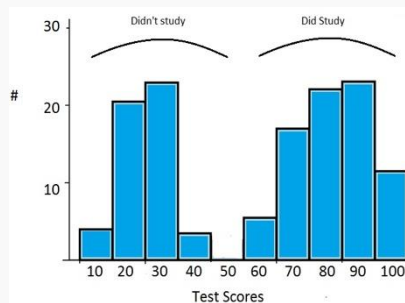
B. Bimodal distribution

3.



C. Symmetric, unimodal distribution with potential outliers

4.

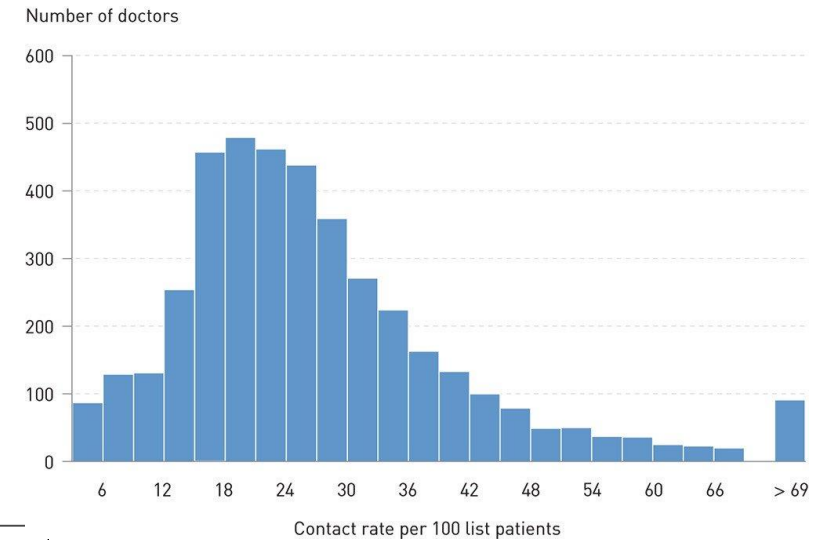


D. Left skewed, unimodal distribution

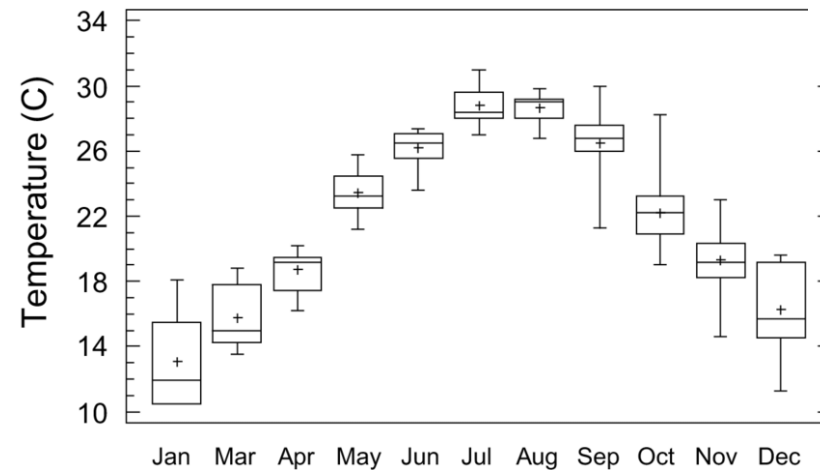
Match each graph with its description

Visually representing data

- Histogram

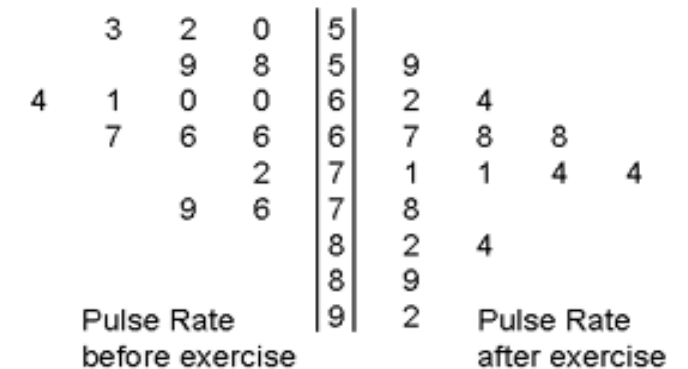


- Box plot



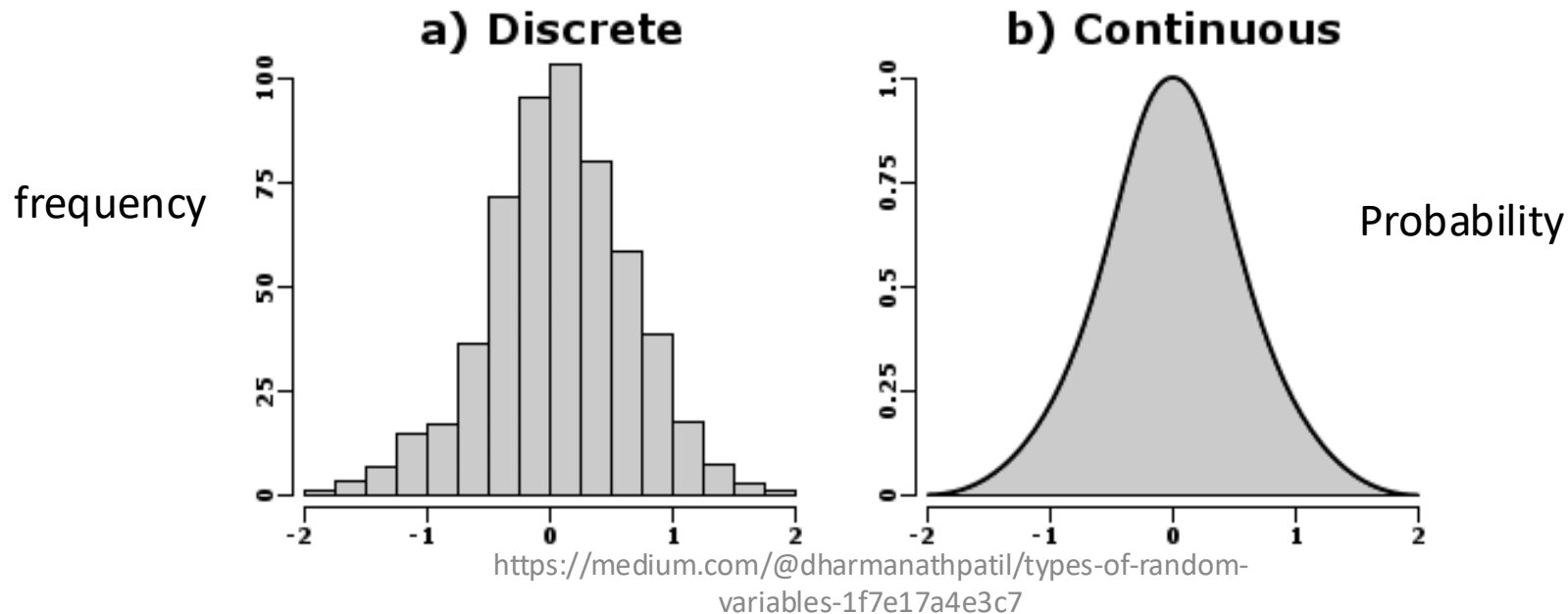
- Stem and leaf

Stem & Leaf Plot



Discrete distributions vs continuous distributions

- **Discrete distribution:** data can only take on certain values, for example integers.
- **Continuous distribution:** data can take on any value within a specified range (which may be infinite).



Discrete or continuous?

- Length of a film?
- Pages in a book?
- Time taken to run a race?
- Number of people in a race?
- Shoe size?
- Temperature?



Discrete and Continuous Data

Discrete data can only take on certain individual values.

Continuous data can take on any value in a certain range.

Example 1

Number of pages in a book is a **discrete variable**.



Example 3

Shoe size is a **Discrete variable**. E.g. 5, $5\frac{1}{2}$, 6, $6\frac{1}{2}$ etc. Not in between.

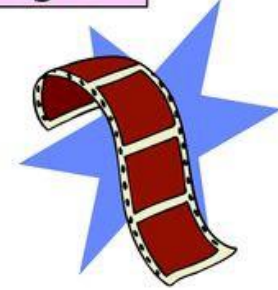


Example 5

Number of people in a race is a **discrete variable**.

Example 2

Length of a film is a **continuous variable**.



Example 4

Temperature is a **continuous variable**.

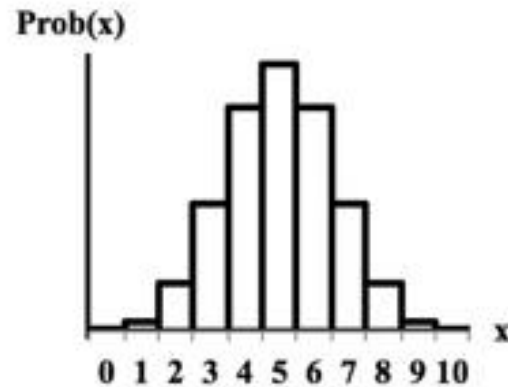
Example 6

Time taken to run a race is a **continuous variable**.

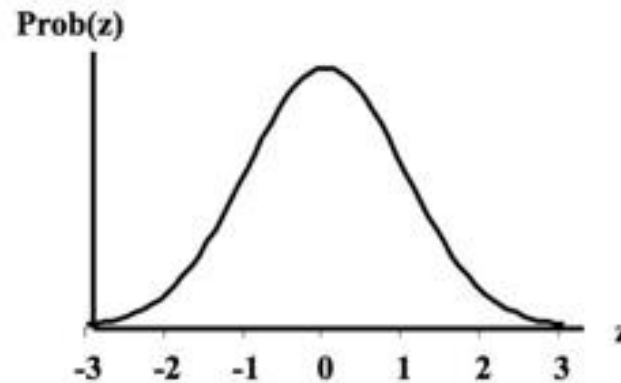


Probability Distributions

A probability distribution is a mathematical function (or just a simple summary) that describes the probability of all possible values of a random variable.



Binomial Distribution
Discrete Data & Discrete
Probability Curve



Standard Normal Distribution
Continuous Data and Continuous
Probability Curve

Distribution	data	Probability Curve
Binomial, Hypergeometric, Poisson	Discrete	Discrete
Exponential, Normal, t	Continuous	Continuous

Bernoulli Trial:

A Bernoulli trial is an experiment with only two possible outcomes

FEATURES of a Bernoulli trial:

- The two possible outcomes are labeled: success (s) and failure (f)
- The probability of success is $P(s)=p$
- The probability of failure is $P(f)=q = 1-p$.
- The probability that something happens (either s or f) $=1=p+q$

Examples:

1. Tossing a coin (success= H , failure= T , and $p=P(H) = 1/2$).
2. Inspecting an item (success=defective, failure=non-defective, and $p=P(\text{defective}) = 1/10000$, say).

Bernoulli Trial: What does success or failure mean?

- There is no universal definition of success or failure
- Definitions are context dependent
- Usually the definition(s) are made by the person conducting the trial

Different people may disagree on the definition for the same situation!

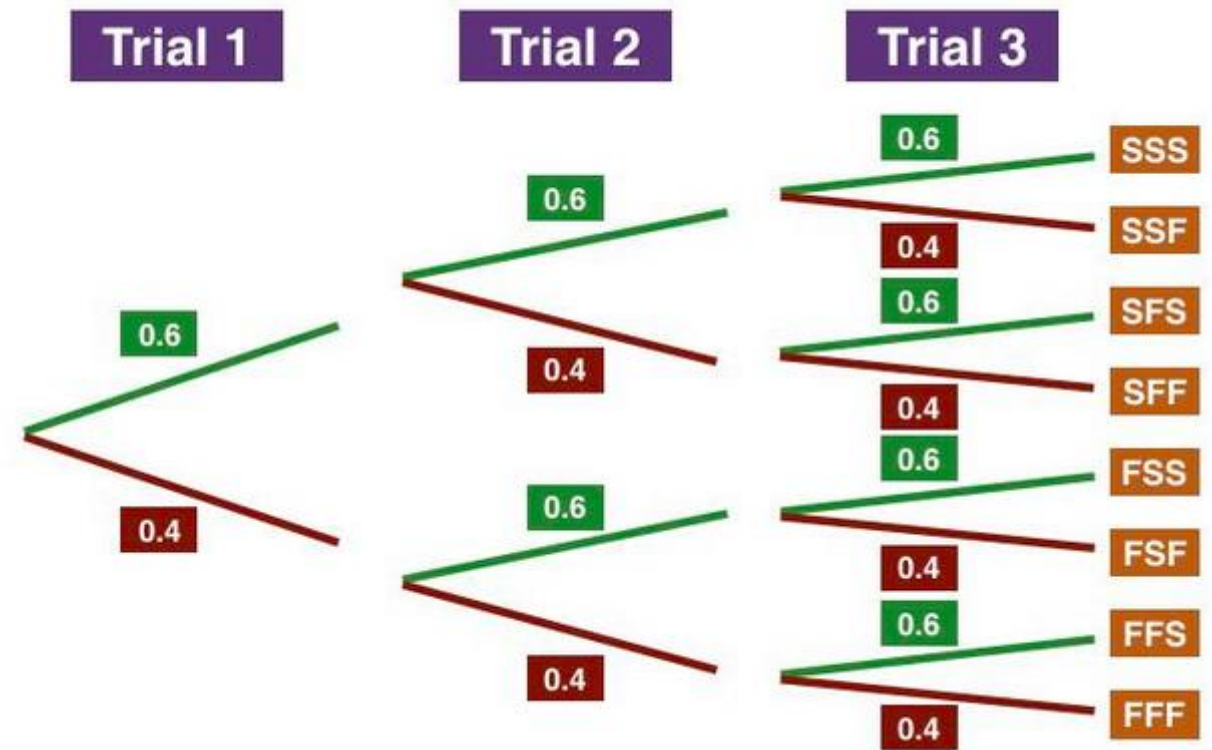
- A factory quality controller might define a success as finding a failure when inspecting items: “I have successfully detected a fault”.
- The factory owner might view the detection of a fault as a failure: “My factory has failed to produce a useable item, I can not sell this, I have lost money”.

Binomial Random Variable:

Consider the random variable:
 X = the number of successes in n identical, Bernoulli trials.

Conditions:

- The probability of success, $P(s)=p$, remains constant from trial to trial.
- The repeated trials are **independent**; that is the outcome of one trial has no effect on the outcome of any other trial.



Example: $p(\text{success}) = 0.6$

Binomial Probability Distribution

- Experiment - **Tossing a coin 2 times (n=2)**
- Outcomes for each trial (two trials): **H = success, T = failure**
- Sample space: **$S = \{HH, HT, TH, TT\}$**
- Random variable: **$X = \text{number of heads}$**

Sample point (Outcome)	Probability	Value of X (x)
HH	$P(HH) = 1/2 \times 1/2 = 1/4$	2
HT	$P(HT) = 1/2 \times 1/2 = 1/4$	1
TH	$P(TH) = 1/2 \times 1/2 = 1/4$	1
TT	$P(TT) = 1/2 \times 1/2 = 1/4$	0

SUMMARY:

- The possible values of X are: 0, 1, and 2.
- X is a discrete random variable.

Define the following events:

Event ($X=x$)	Probability = $P(X=x)$
$(X=0)=\{TT\}$	$P(X=0) = P(TT) = 1/4$
$(X=1)=\{HT, TH\}$	$P(X=1) = P(HT) + P(TH) = 1/2$
$(X=2)=\{HH\}$	$P(X=2) = P(HH) = 1/4$

In a table, show the possible values of X with their probabilities:

X	0	1	2	Total
$P(X=x)$	$1/4$	$1/2$	$1/4$	1.00

Binomial Distribution : main points / features

- A statistical tool used to find the probability of the number of successes that are likely to occur in a sequence of independent trials
- BINS (Conditions for use that must be met!)
 - Binary : trials can only have 2 possible outcomes (success/failure; yes/no; true/false)
 - Independent : Outcome of any given trial has no effect on subsequent trials
 - Number : there must be a fixed number of trials
 - Same : the probability of success / failure must be the same (fixed) for all trials
- Let N = number of trials; p = probability of success ; $q = 1 - p$ = probability of failure then
 - Mean : expected number of successes in N trials is $\mu = Np$
 - Variance : measure of spread of the distribution is $\sigma^2 = Np(1 - p)$
- When N is very large the Binomial distribution is well approximated by the Normal distribution (the subject of the week 7 lecture)

Binomial Probability Distribution

https://onlinestatbook.com/2/calculators/binomial_dist.html

Online calculator for graph of binomial probability distribution.

N =number of trials

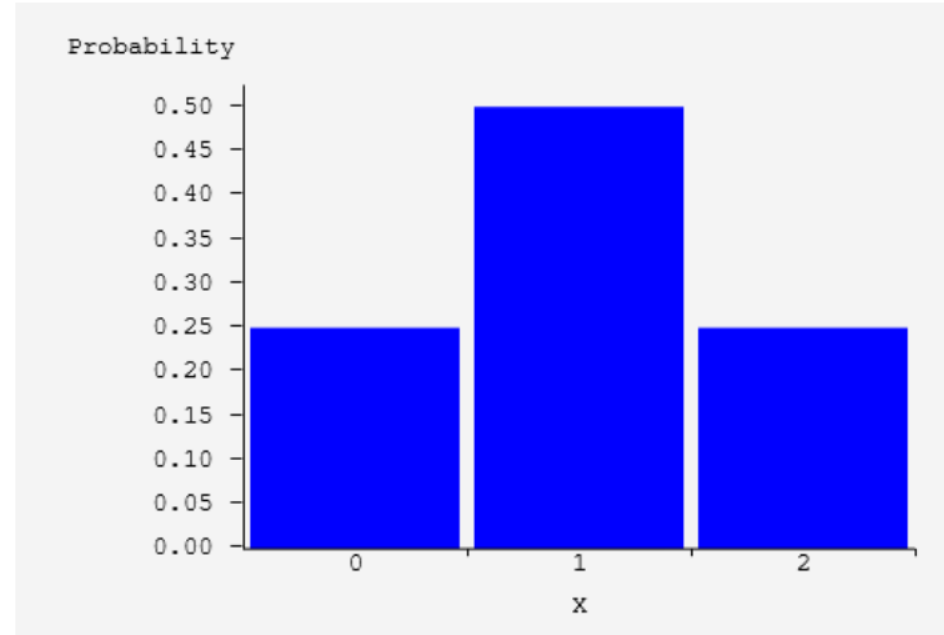
π =probability for a success

Expand the example
to tossing a coin 3 times:

$n=3$

<https://www.mathsisfun.com/data/binomial-distribution.htm>

Binomial Distribution



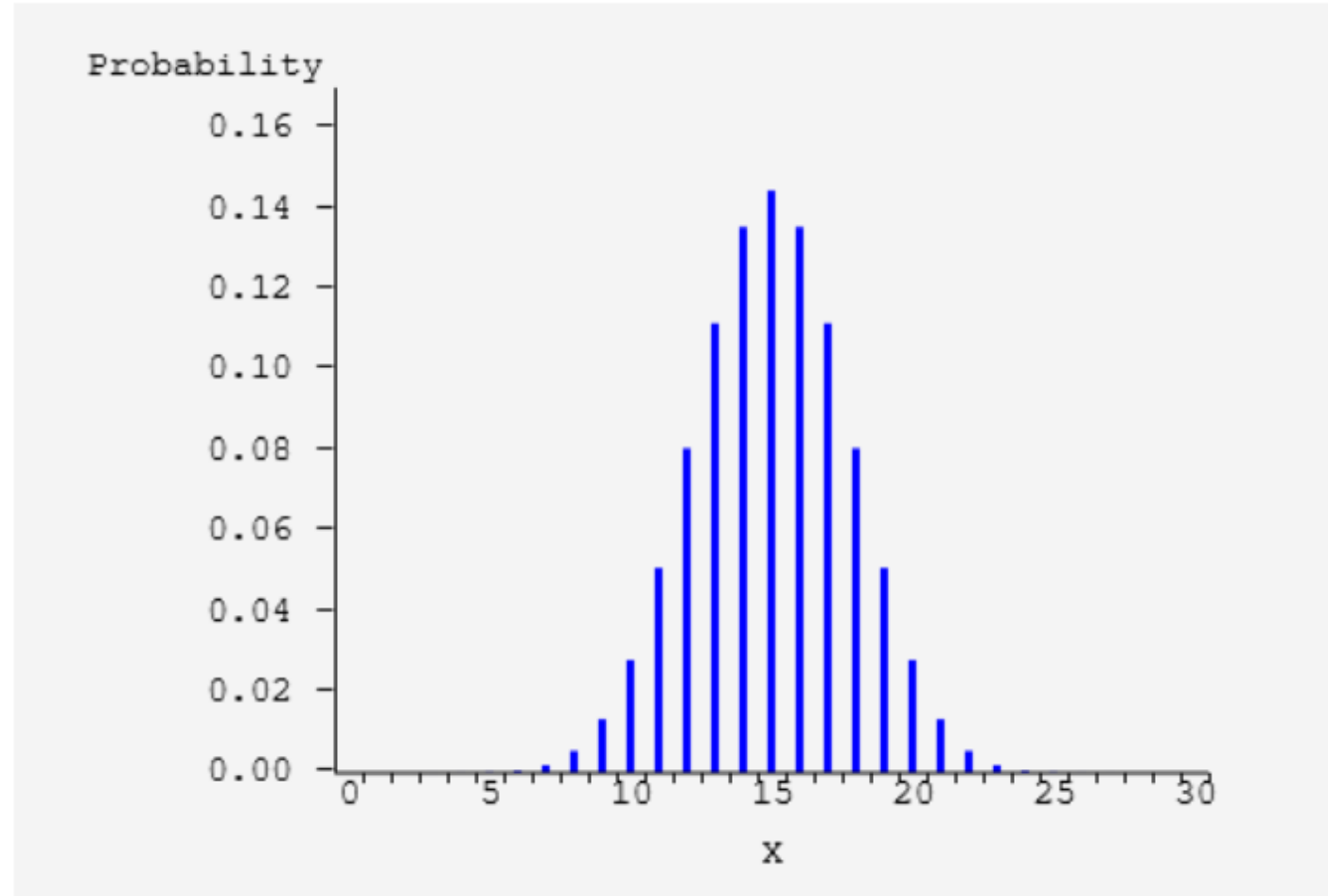
N
 π

Binomial Distribution

This shows the probability distribution of flipping (x) heads in 30 flips of a coin.

This one looks like a normal distribution but with gaps ...

Binomial Distribution is DISCRETE



N
Π

Binomial distributions

Problem: Determine whether the given scenario describes a binomial distribution for X . Justify your answer.

Here X = number of children with type O Blood

(a) Genetics says that the genes children receive from their parents are independent from one child to another. Each child of a particular set of parents has probability 0.25 of having type O blood. Suppose these parents have 5 children. Count the number of children with type O blood.

1. There are only 2 possible outcomes for each trial (binary)
2. There are a fixed number, n , of IDENTICAL trials
3. The probability of success ' p ' at each trial is consistent
4. Each trial is INDEPENDENT

Binomial distributions

Problem: Determine whether the given scenario describes a binomial distribution for X . Justify your answer.

Here X = number of children with type O Blood

(a) Genetics says that the genes children receive from their parents are independent from one child to another. Each child of a particular set of parents has probability 0.25 of having type O blood. Suppose these parents have 5 children. Count the number of children with type O blood.

(a)

- **Binary?** “Success” child has type O blood. “Failure” child doesn’t have type O blood.
- **Number?** $n = 5$
- **Same probability?** $p = 0.25$
- **Independent?** Knowing one child’s blood type tells you nothing about another child’s because they inherit genes independently from their parents.

Yes, X = number of children with type O is a binomial random variable

Binomial random variables

Problem: Determine whether the given scenario describes a binomial distribution for Y . Justify your answer.

(b) Shuffle a standard deck of 52 playing cards. Turn over the first 10 cards, one at a time. Record the number of aces you observe. That is, out of the 10 cards, Y = number of aces.

Binomial random variables

Problem: Determine whether the given scenario describes a binomial distribution for Y . Justify your answer.

(b) Shuffle a standard deck of 52 playing cards. Turn over the first 10 cards, one at a time. Record the number of aces you observe. That is, out of the 10 cards, Y = number of aces.

(b)

- **Binary?** “Success” is get an ace. “Failure” is don’t get an ace.
- **Independent?** No. If the first card you turn over is an ace, then the next card is less likely to be an ace because you’re not replacing the top card in the deck. If the first card isn’t an ace, the second card is more likely to be an ace.

This is not a binomial random variable because

- 1) The probability of getting an ace is not constant at each trial.
- 2) the independent condition is not met.

1. There are only 2 possible outcomes for each trial (Binary)
2. There are a fixed number, n , of IDENTICAL trials
3. The probability of success ' p ' at each trial is consistent
4. Each trial is INDEPENDENT



Try another example:

If 75% of drivers pass their driving test, what is the probability that 3 of the next 5 pass?

First is this binomial?

1. Yes, only 2 possible outcomes pass or fail.
2. Yes, fixed number of trials (5) $n=5$
3. Probability of success is consistent $p = 0.75$
4. Each trial is independent (test does not care if last person passed or failed)

If 75% of drivers pass their driving test,
what is the probability that 3 of the next 5 pass?



We could try:

$$\begin{aligned} P(\text{SSSFF}) &= 0.75 \times 0.75 \times 0.75 \times 0.25 \times 0.25 \\ &= (0.75)^3 \times (0.25)^2 \end{aligned}$$

But this does not take into consideration order of drivers.
We need to multiply by the number of ways of selecting
3 drivers out of 5:

$$\begin{aligned} P(3 \text{ pass}) &= {}^5C_3 \times (0.75)^3 (0.25)^2 & {}^5C_3 &= 10 \\ &= 0.2637 \text{ (4dp)} \end{aligned}$$

$$P(3 \text{ pass}) = {}^5C_3 \times (0.75)^3 \times (0.25)^2 = 0.2637 \text{ (4dp)}$$

If: n = the number of trials

x = the number of successes

p = the probability of success in one trial

$q = 1-p$ = the probability of failure in one trial

Then the probability of x successes in n trials is given by:

$$P(X = x) = P(x \text{ successes}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$${}^nC_x \rightarrow = {}^nC_x p^x q^{n-x}$$

Binomial Probability Distribution

The Binomial Distribution for the variable X describes the probability of x successes out of a fixed number of independent trials.

If: n = the number of trials

p = the probability of success in one trial

Then the probability of x successes in n trials is given by:

$$P(X = x) = P(x \text{ successes}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

${}^n C_x$ 

Combinations review

Remember

$${}^5C_3 = \frac{5!}{(5-3)!3!} = 10$$

Combination Formula

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

Combinations and Pascal's triangle

Now that you are familiar with combinations, there is another important pattern that you can recognize. Each term in Pascal's triangle corresponds to a value of ${}_nC_r$.

																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					</
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	----

											${}_0C_0$
							${}_1C_0$		${}_1C_1$		
					${}_2C_0$		${}_2C_1$		${}_2C_2$		
			${}_3C_0$		${}_3C_1$		${}_3C_2$		${}_3C_3$		
		${}_4C_0$		${}_4C_1$		${}_4C_2$		${}_4C_3$		${}_4C_4$	
${}_5C_0$		${}_5C_1$		${}_5C_2$		${}_5C_3$		${}_5C_4$		${}_5C_5$	

Driving Example done again:



© dreamstime.com

ID 157591158 © ClassicVector

If 75% of drivers pass their driving test,
what is the probability that 3 of the next 5 drivers pass?

Parameters:

$$P(x \text{ successes}) = {}^nC_x p^x (q)^{n-x}$$

$$n = 5$$

$$x = 3 \text{ (want 3 successes)}$$

$$P(X = 3) = {}^5C_3 (0.75)^3 (0.25)^{5-3}$$

$$p = 0.75 \text{ success}$$

$$q = 0.25$$

$$\begin{aligned} P(X = 3) &= (10) \times (0.75)^3 \times (0.25)^2 \\ &= 0.2637 \end{aligned}$$

Graphics Calculator emulator free download

<https://www.intereseducation.com/resources/casio-fx-9860g-sd-graphing-calculator-emulator-full-version/>

Online Binomial Distribution Calculator

https://onlinestatbook.com/2/calculators/binomial_dist.html

Driving Example done again:

If 75% of drivers pass their driving test, what is the probability that 3 of the next 5 drivers pass?

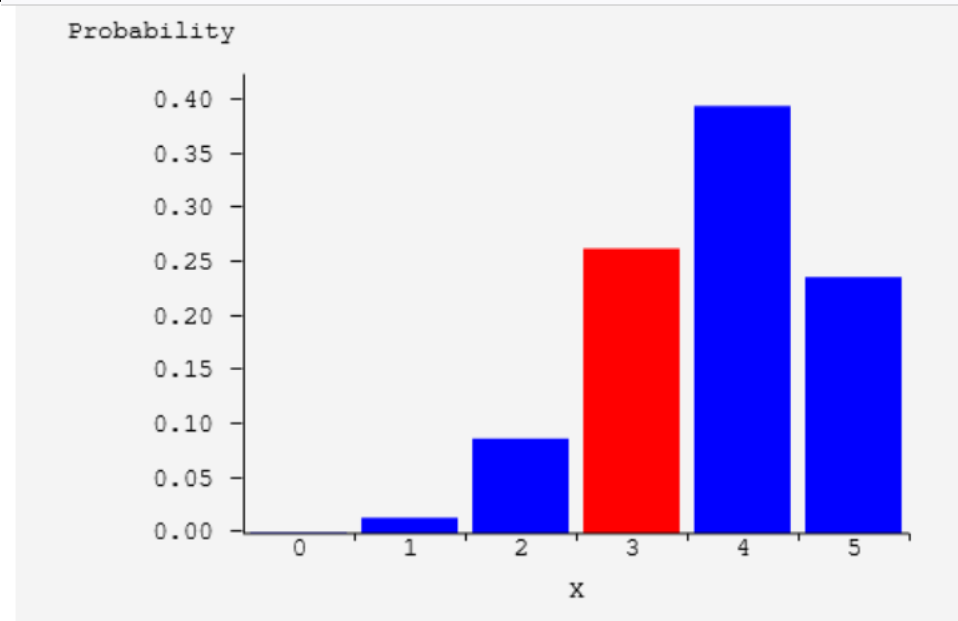
Parameters:

$n = 5$

$x = 3$ (want 3 successes)

$p = 0.75$ success

$q = 0.25$



N
 π

☐ Above
☐ Below
☒ Between and inclusive

Probability = 0.2637

Watch for inequalities!!!

$$P(X < 2) = P(X \leq 1) \quad 0, 1, 2, 3, 4, 5, 6, \dots$$

$$P(X > 2) = 1 - P(X \leq 2) \quad 0, 1, 2, 3, 4, 5, 6, \dots$$

$$P(X \geq 2) = 1 - P(X \leq 1) \quad 0, 1, 2, 3, 4, 5, 6, \dots$$

Using R to compute binomial probabilities

A) Function `dbinom(k, n, p)` gives **single**
binomial probabilities

`dbinom(k, n, p)` gives $P(X = k)$ with n and p fixed

k = number of successes

n = Number of trials

p = probability of success

Using R to compute binomial probabilities

A) Function `dbinom(k, n, p)` gives **single**
binomial probabilities

Examples: $X \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

`dbinom(6, 10, 0.3)` gives $P(X = 6)$ where $X \sim \text{Binomial}(10, 0.3)$

`dbinom(2:4, 10, 0.3)` gives $P(X = 2)$, $P(X = 3)$, and $P(X = 4)$
where $X \sim \text{Binomial}(10, 0.3)$

`dbinom(17, 10, 0.3)` returns 0 ! Careful, remember $k \leq n$.

Using R to compute binomial probabilities

B) Function `pbinom(k, n, p)` gives **cumulative**
binomial probabilities

`pbinom(k, n, p)` gives $P(X \leq k)$ with n and p fixed

k = number of successes

n = Number of trials

p = probability of success

Using R to compute binomial probabilities

B) Function `pbinom(k, n, p)` gives **cumulative**
binomial probabilities

Examples: $X \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

`pbinom(6, 10, 0.3)` gives $P(X \leq 6)$ where $X \sim \text{Binomial}(10, 0.3)$

i.e., $P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6)$!!

OUTPUT:

```
> pbinom(6, 10, 0.3)
[1] 0.9894079
```

Example:

If the probability of passing drivers test is 75%,
what is the probability of **at most 2** out of next 5
passing?

$P(X=0)$ or $P(X=1)$ or $P(X=2)$.

This can be written $P(X<3)$ or $P(X\leq 2)$.

$$P(X\leq 2) = P(X=0) + P(X=1) + P(X=2) \quad = 0.1035$$

Example:

Batteries are supplied in boxes of 144. The probability that a battery is faulty is 5%.

- a) What is the probability that less than 10 batteries are faulty?
- b) What is the probability that at least 8 batteries are faulty?
- c) What is the probability that between 5 and 15 batteries are faulty?

$$n = 144, \quad p = 0.05$$

$$\begin{aligned} a) P(x < 10) &= P(x \leq 9) \\ &= (0.81465) \end{aligned}$$

$$\begin{aligned} b) P(x \geq 8) &= 1 - P(x \leq 7) \\ &= 1 - 0.5682 \\ &= 0.4318 \end{aligned}$$

$$\begin{aligned} c) P(5 \leq x \leq 15) &= P(x \leq 15) - P(x \leq 4) \\ &= 0.9976 - 0.1487 \\ &= 0.8489 \end{aligned}$$

R studio commands

dbinom(r,n,p) gives the probability of r successes in n trials:

$$P(X=r)$$

pbinom(r,n,p) gives the cumulative probability of **at most** r successes in n trials:

$$P(X \leq r)$$

R studio commands

`dbinom(r,n,p)` gives the probability of r successes in n trials: $P(X=r)$

`pbinom(r,n,p)` gives the cumulative probability of at most r successes in n trials: $P(X \leq r)$

Batteries are supplied in boxes of 144. The probability that a battery is faulty is 5%.

a) What is the probability that less than 10 batteries are faulty?

$$P(x < 10) = P(x \leq 9) = \text{pbinom}(9,144,0.05)$$

b) What is the probability that at least 8 batteries are faulty?

$$P(x \geq 8) = 1 - P(x \leq 7) = 1 - \text{pbinom}(7,144,0.05)$$

c) What is the probability that between 5 and 15 batteries are faulty?

$$P(5 \leq x \leq 15) = P(x \leq 15) - P(x \leq 4) = \text{pbinom}(15,144,0.05) - \text{pbinom}(4,144,0.05)$$

Expected Value

The expected value $E(X) = \mu$ is the value we would get if we did a long run experiment and averaged the results

** Expected value of a random variable = mean of a random variable

FOR DISCRETE RANDOM VARIABLES:

$$\mu = E(x) = \sum xp(x)$$

x	1	2	3	4	5	6
P(X=x)	1/6	1/6	1/6	1/6	1/6	1/6

$$\begin{aligned} E(X) &= \frac{1}{6} \times 1 + \frac{1}{6} \times 2 + \frac{1}{6} \times 3 + \frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6 \\ &= 3.5 \end{aligned}$$

Expected Value

X	x_1	x_2	x_3	x_4		x_n
$P(X=x)$	$P(x_1)$	$P(x_2)$	$P(x_3)$	$P(x_4)$		$P(x_n)$

$$E(X) = x_1p(x_1) + x_2p(x_2) + \dots + x_np(x_n)$$

$$E(X) = \sum_{i=1}^{i=n} x_i p(x_i)$$

Example

x	1	2	3	4	5
P(X=x)	0.3	0.1	0.2	0.1	0.3

$$\begin{aligned} E(X) &= 1(0.3) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.3) \\ &= 3 \end{aligned}$$

Applications - I

The expected value has applications especially in the insurance industry, and gambling

An insurance company sells term life insurance policies for a 20-year period. The amount payable if death occurs is \$75,000. The probability of the death of a 30-year old man over the next 20 years is 0.017. The insurance premium is \$2000. What is the company's expected gain?

The gain to the insurance company is the random variable with two possible values

\$2000 gain = premium

\$75000-2000= payout

x	2000	-73000
P(X=x)	0.983	0.017

$$\begin{aligned}E(X) &= 2000 \times 0.983 + (-73000) \times 0.017 \\ &= \$725\end{aligned}$$

Applications - II

A lottery has 3 prizes: \$2000, \$500, \$100. If there are 1300 tickets sold, what should the organizer charge for each ticket for the lottery to be fair (gain is zero)?

Assume ticket price = p

Consider the possibilities: win $\$2000-p$, the ticket price

win $\$500-p$, the ticket price

win $\$100-p$, the ticket price

lose $-p$, cost of ticket

x	$2000-p$	$500-p$	$100-p$	$-p$
$P(X=x)$	$\frac{1}{1300}$	$\frac{1}{1300}$	$\frac{1}{1300}$	$\frac{1297}{1300}$

$$E(X) = (2000 - p) \times \frac{1}{1300} + (500 - p) \times \frac{1}{1300} + (100 - p) \times \frac{1}{1300} + (-p) \times \frac{1297}{1300}$$

$$E(X) = (2000 - p) \times \frac{1}{1300} + (500 - p) \times \frac{1}{1300} + (100 - p) \times \frac{1}{1300} + (-p) \times \frac{1297}{1300}$$

If the lottery is fair then the expected gain is zero

$$(2000 - p) \times \frac{1}{1300} + (500 - p) \times \frac{1}{1300} + (100 - p) \times \frac{1}{1300} + (-p) \times \frac{1297}{1300} = 0$$

$$\frac{(2000 - p) + (500 - p) + (100 - p) - 1297p}{1300} = 0$$

$$\frac{2600 - 1300p}{1300} = 0$$

$$2600 - 1300p = 0$$

$$1300(2 - p) = 0$$

$$P = \$2$$

The price of the ticket should be \$2

Expected Value in Rstudio

- <https://www.youtube.com/watch?v=yuhRgelGhKA>
'Expected Value in R' – watch first 1.18 minutes.

Expected Value and Variance of a Binomial Distribution

- <https://www.youtube.com/watch?v=beGYOZppbz4>

'R Tutorial: Expected value and variance'

What did you get out of today?

- What was the purpose of today's lecture?
- What is one key thing you learnt?
- Is there anything that puzzled me?